

Chiplet-LEGO: Delivering Multiple Voltage Rails to Chiplets with Chiplet VRMs

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Abstract—This paper presents Chiplet-LEGO, a single-input multiple-output (SIMO) linear extendable group operated (LEGO) voltage regulator module (VRM) architecture as a new development of the LEGO-VRM family. This architecture not only offers a high voltage conversion ratio but also delivers multiple individually regulated output voltages, making it well-suited for VRMs in system-on-chip (SoC) applications with multiple voltage rails. The Chiplet-LEGO is well suited to numerous numbers of independently regulated output voltages with scalable power capability. Dynamic models were derived to understand the cross-regulation behaviors. A 24 V-to-1 V 120 W dual-output Chiplet-LEGO prototype with discrete components was built and tested to evaluate the effectiveness of the analysis.

Index Terms—Chiplet-VRM, Intermediate-voltage-bus, Modularity, Multiple outputs, Cross regulation, Dynamic model

I. INTRODUCTION

Artificial intelligence (AI) is a pivotal driver of technological advancement. Numerous AI systems are emerging, necessitating high-performance power management units (PMUs) with smart power delivery [1], [2]. Modern microprocessors and accelerators in computing systems are becoming increasingly power-hungry, making power management a critical bottleneck. To improve energy efficiency, processors incorporate multiple voltage domains with dynamic voltage and frequency scaling (DVFS), enabling different power supply strategies tailored to multiple independent workloads [3]. Power management units (PMUs) that provide multiple voltage domains, meet transient response requirements, and show robustness against disturbances are needed [4]–[8].

LEGO-VRM is a family of power delivery architecture developed for high current computing systems [9], [10]. It offers scalable output power, structural flexibility (enabling vertical power delivery), high efficiency, and high power density. As illustrated in Fig. 1, the CPUs, GPUs, and NPUs may be fabricated using different process nodes and be packaged together, necessitating distinct operating voltage domains. Heterogeneous integration allows computing systems, memory systems, and communication systems being packaged in close proximity. To address these requirements, this paper presents the Chiplet-LEGO architecture as a new development of the LEGO-VRM family. Chiplet-LEGO inherits the advantages of the LEGO-VRM family, such as voltage balancing, current sharing, and soft charging, while providing multiple output voltage rails, creating a large design trade-off margin among efficiency, power density, and transient speed [2].

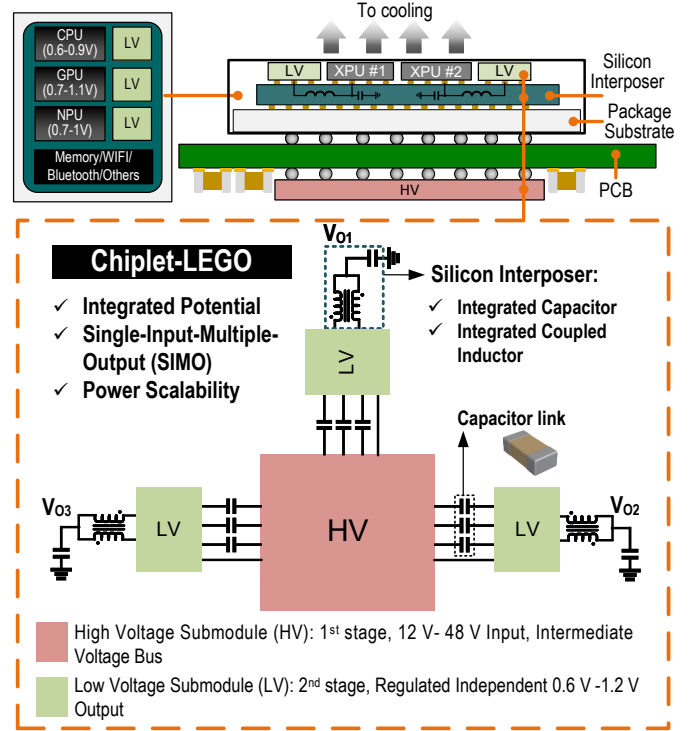


Fig. 1. Principles of the Chiplet-LEGO VRM architecture for vertical power delivery (VPD). The Chiplet-LEGO splits into different power submodules, each with distinct functions with corresponding passive components.

A prototype dual-output 24 V-to-1 V Chiplet-LEGO achieved 85.8% peak power efficiency at a rated output power of 56 W. The two output rails can be independently changed between 0.5 V to 1.2 V each delivering 55 A. The rated power of the full system is 120 W. The switching frequency of the voltage regulation stage is 1.266 MHz.

II. OPERATING PRINCIPLES OF CHIPLET-LEGO VRM

Linear extendable group operated point-of-load (LEGO-PoL) architecture is a merged two-stage hybrid-switched-capacitor architecture that can be optimized with 3D stacked packaging and coupled inductors to achieve miniaturized size and high power efficiency [9], [10]. Figure 2 shows the key principles of the Chiplet-LEGO architecture with N 2:1 switch capacitor unit and M buck converter unit. These M buck stages are independent from each other. They don't need to be operated with interleaving. Note that the output of each of

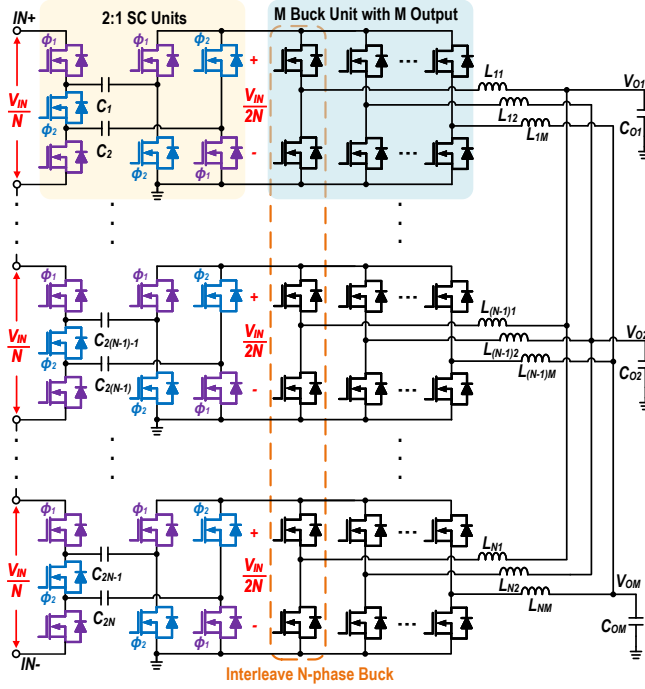


Fig. 2. N -submodules and M -phase buck converter of the Chiplet-LEGO architecture with M output voltage rails. One LEGO-PoL submodule comprises one 2:1 switched-capacitor unit and one M -phase buck with independent output unit. N submodules are connected with input in series and output in parallel. The Chiplet-LEGO PoL architecture can be linearly extended for different input voltage, output current ranges and output voltage number.

these M buck converter comprises equal numbers of single-phase buck units from each voltage domain to balance the charge and discharge of the intermediate voltage bus and the flying capacitors.

The 2:1 switched capacitor unit operates with fixed complementary 50% duty cycles (Φ_1 and Φ_2) with lower switching frequency to reduce the switching loss at the cost of larger capacitor voltage ripple. The buck converters operate at a higher switching frequency for high control bandwidth. There is no large dc decoupling capacitor between the two stages. In practice, it is beneficial to include decoupling capacitors to compensate for the loop inductances [9]. The two stages are functionally merged: the buck units act as current sources to soft-charge the switched capacitors. The Chiplet-LEGO architecture is modular and scalable. One can always parallel more buck units if more output current is desired. The voltage balancing and current sharing of the Chiplet-LEGO architecture inherit those of the LEGO-VRM family.

III. DESIGN OF A 24 V-TO-1 V DUAL-OUTPUT CHIPLET-LEGO VRM

To illustrate the principles of the Chiplet-LEGO architecture, this paper presents the design of a 24 V-to-1 V dual-output Chiplet-LEGO VRM with 2 switched-capacitor (SC) units and 2 interleave buck converters. Fig. 3 (a) shows the Chiplet-LEGO circuit topology contains dual-channel MOSFETs and DrMOSes in the SC stage, small intermediate bus filter capacitor C_{f1} and C_{f2} , and 4 buck stage DrMOS. The switching frequency relationship between SC stage and buck

stage follows (1) [9]. In this design, f_{sc} is 281.3 kHz and $k = 4$.

$$f_{buck} = \frac{2k+1}{2} f_{sc}, k = 1, 2, 3, \dots \quad (1)$$

where f_{buck} is 1.266 MHz, the buck converter switching frequency. f_{sc} is SC stage switching frequency. k is an integer number. Note that the switching frequency can be different between these two 2-phase buck converters.

Figure 3 (b) illustrates the steady-state operation waveforms of the Chiplet-LEGO. Φ_1 and Φ_2 are complementary SC gate signals. D_1 and D_2 refer to the buck converter duty ratio with respect to V_{OUT1} and V_{OUT2} . V_{B11} and V_{B21} share the same duty ratio and interleave (180° phase shift) to each other. To achieve zero-current-switching (ZCS), the buck gate signals do not overlap with the SC stage gate signals when they flip.

A. Experimental Results

To validate the design and performance of the Chiplet-LEGO converter, a 24 V-to-1 V dual-output Chiplet-LEGO prototype was constructed. The floating switches in the switched capacitor (SC) stage are implemented with Onsemi NTTFD2D8N03P1E dual-channel MOSFETs, which are driven by Infineon 2EDF7275K isolated gate driver with TI UCC27212 charge pump for bootstrapping. Other switches in the SC stage are implemented with Onsemi FDMF3039 DrMOS, and all the switches in the buck stage are implemented with Infineon TDA21490 DrMOS. Each buck DrMOS is followed by a Coilcraft SLC7649S-101KL 100 nH inductor. Each flying capacitor value are 40 μ F ($12 \times 3.3\mu$ F) with 0805 footprints, and the filter capacitor value is 3.3 μ F. Fig. 4(a) exhibits the PCB design of the dual-output Chiplet-LEGO prototype. Since this design aims to verify the concept of multiple-output, it is not in a vertical power delivery layout for maximum power efficiency and power density optimization purpose. All power switches and drivers are located on the front side and flying capacitors are on the back side of the board as shown in Fig. 4 (a). Fig. 4 presents measured V_{BUS1} , V_{B11} , V_{B12} , and V_{SC} signals under steady-state when $V_{OUT1} = 0.6$ V ($D_1 = 0.13$), $V_{OUT2} = 0.9$ V ($D_2 = 0.18$) and $I_{OUT1} = I_{OUT2} = 5$ A. Where V_{B11} and V_{B12} are switching nodes of the buck converters and V_{SC} is one of the gate signals of SC stage Dual Channel MOSFETs.

The measured efficiency is shown in Fig. 5 (a). The blue curve indicates single-output situation ($V_{OUT1} = 1$ V and $V_{OUT2} = 0$ V. Here V_{OUT2} related buck DrMOS devices are not switching), achieving 88% peak efficiency when $I_{OUT1} = 15$ A. The red curve indicates dual-output situation ($V_{OUT1} = 1$ V and $V_{OUT2} = 0.6$ V), achieving 85.8% peak efficiency. Notice that I_{OUT2} is set as a constant current which is 10 A. Compared to these two scenarios, the single-output case achieves higher peak efficiency due to lower switching losses. However, the dual-output case delivers better efficiency at higher total output power. Under heavy load conditions, conduction losses dominate the efficiency, and the second output shares the total output power, thereby reducing the inductor current. Fig. 5 (b) presents the heat distribution when $V_{OUT1} = 1$ V, $V_{OUT2} = 0.6$ V and total power is 56 W.

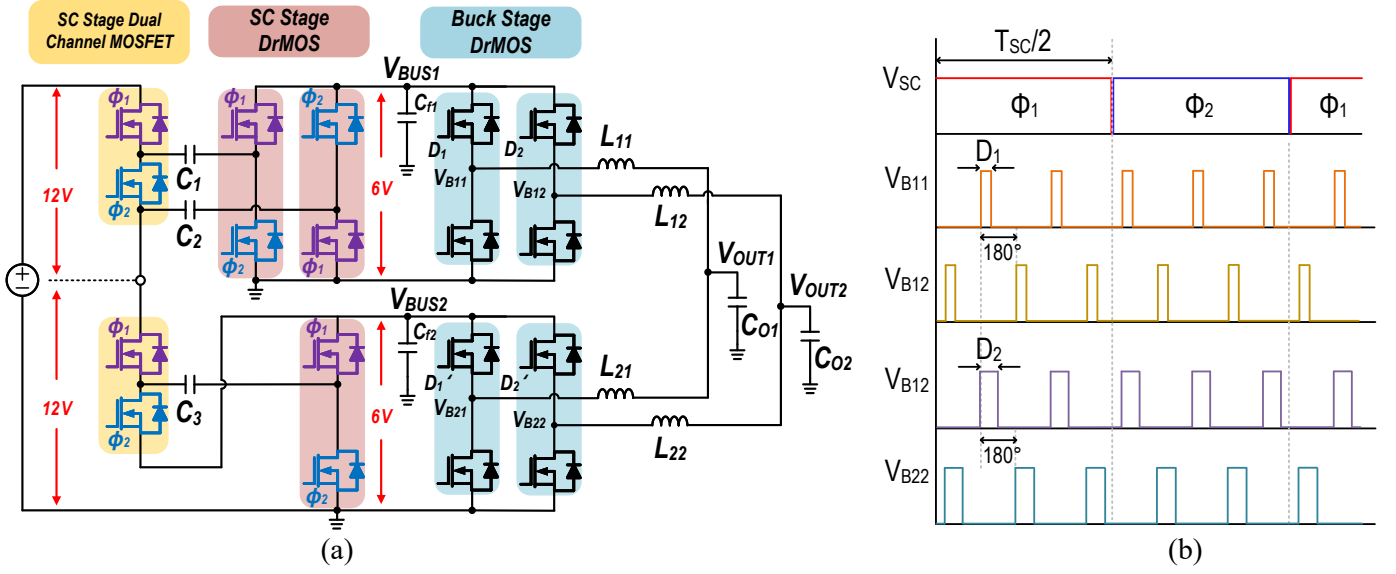


Fig. 3. (a) 24 V-to-1 V dual-output Chiplet-LEGO VRM topology schematic. There are two intermediate bus voltage V_{BUS1} and V_{BUS2} . (b) Operation waveforms of the dual-output Chiplet-LEGO VRM. Each output draws equal current from all voltage domains.

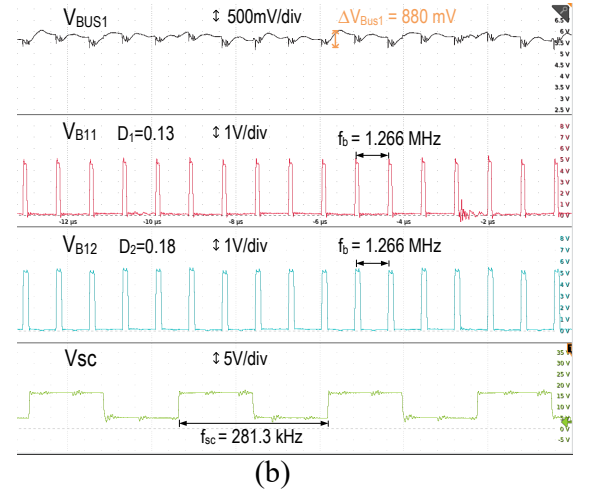
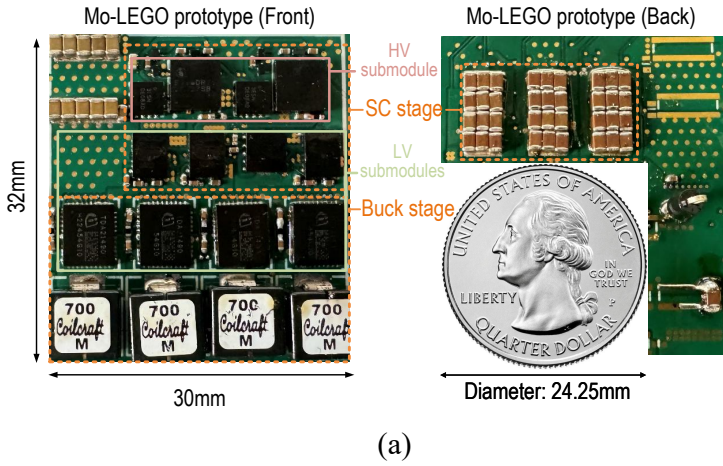


Fig. 4. (a) The active PCB area of the 24 V-to-1 V dual-output Chiplet-LEGO. (b) Measured steady-state key operation waveforms. V_{SC} is one of the gate signal of SC stage Dual Channel MOSFET. A 880 mV ripple was observed on the 6 V virtual intermediate bus.

As expected, the buck DrMOS devices experience the highest temperature under heavy load conditions.

IV. CONTROL DYNAMICS OF CHIPLET-LEGO

Chiplet-LEGO is a typical hybrid-switched capacitor circuit with many L - C components, leading to multi-resonant dynamics [11], [12] and balancing issues [4], [13], [14]. To obtain the dynamic model of the 2SC + 4Buck converter, we begin with the simplest configuration, the 1SC + 1Buck converter, and progressively extend the model to the full 2SC + 4Buck topology, as illustrated in Fig. 6. We started from modifying the state-space averaging (SSA) method [15]–[17], then investigated the dynamic behavior induced by variations in the intermediate bus voltage. In the 1SC + 2Buck topology, we derived the cross-regulation transfer function. Finally, the complete dynamic behavior of the 2SC + 4Buck dual-output

Chiplet-LEGO architecture is established. Similar modeling and analysis methods are applicable to other multi-output hybrid switched capacitor circuits.

A. 1SC + 1Buck: Modified Averaging Method

The simplest Chiplet-LEGO architecture employs a merged 2:1 switched-capacitor (SC) converter at its front end, followed by a single buck converter at its back end. The SC front-end operates with slower switching frequency (f_{sc}) while the buck converter operates with faster switching frequency (f_b) to ensure good efficiency and fast transient response.

We begin by analyzing the 1SC + 1Buck converter using the classic SSA approach. Assuming state vector $x = [v_O, v_{cf}, i]^T$ and the input vector $u = [v_{in}]$, the state equation is

$$\dot{x} = Ax + Bu \quad (2)$$

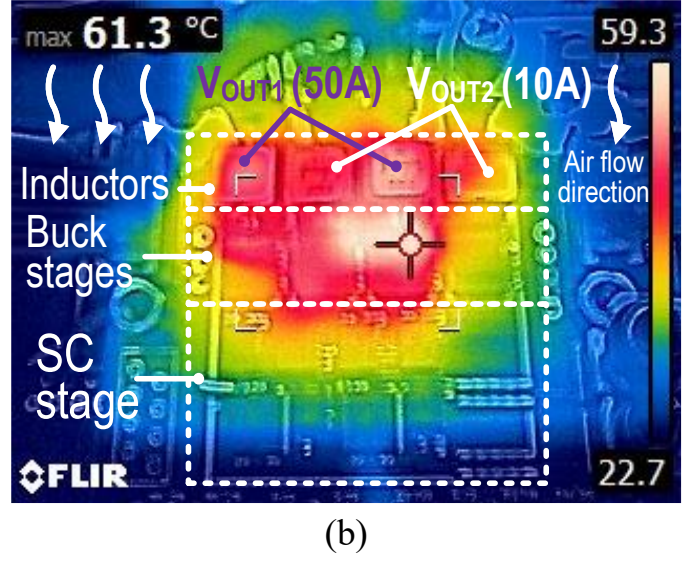
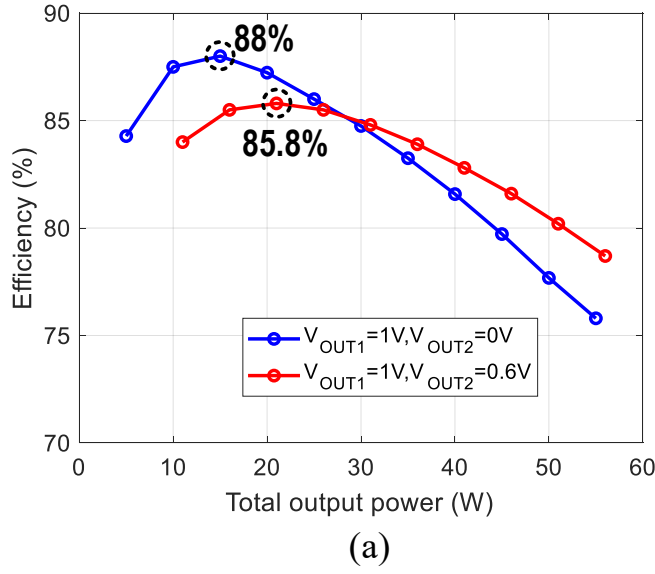


Fig. 5. (a) Measured efficiency versus total output power when $V_{IN} = 24$ V. Blue curve refers to $V_{OUT1} = 1$ V and $V_{OUT2} = 0$ V, red curve refers to $V_{OUT1} = 1$ V and $V_{OUT2} = 0.6$ V. (b) Thermal performance under two DC fans with maximum speed of 26500 RPM when $V_{OUT1} = 1$ V and $V_{OUT2} = 0.6$ V. The buck DrMOS experience highest temperature, as expected.

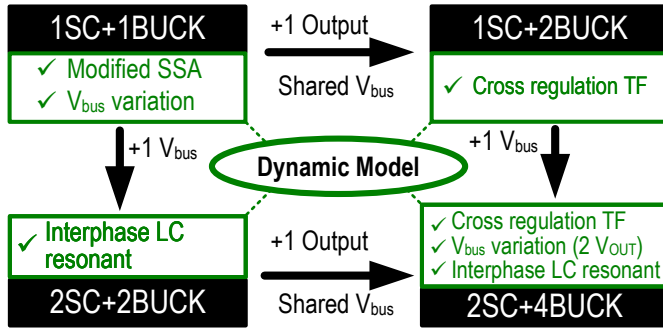


Fig. 6. A variety of different implementations of the Chiplet-LEGO architecture ranging from “1SC + 1Buck” to “2SC + 4Buck” step by step.

The state equations during D in Φ_1 as shown in Fig. 7 (b) are:

$$\begin{aligned} C \frac{dv_O}{dt} &= i - \frac{V_O}{R} \\ C_f \frac{dv_{cf}}{dt} &= i \\ L \frac{di}{dt} &= v_{in} - v_{cf} - v_O \end{aligned} \quad (3)$$

during $1 - D$ in Φ_1 :

$$\begin{aligned} C \frac{dv_O}{dt} &= i - \frac{V_O}{R} \\ L \frac{di}{dt} &= -v_O \end{aligned} \quad (4)$$

Rewrite the equations in matrix format based on (3) and (4), one can obtain the state matrix A_1 , A'_1 , B_1 , and B'_1 respectively:

$$A_1 = \begin{bmatrix} \frac{-1}{RC} & 0 & \frac{1}{C} \\ 0 & 0 & \frac{1}{C_f} \\ \frac{-1}{L} & \frac{-1}{L} & 0 \end{bmatrix}, \quad B_1 = \begin{bmatrix} 0 \\ 0 \\ \frac{1}{L} \end{bmatrix}$$

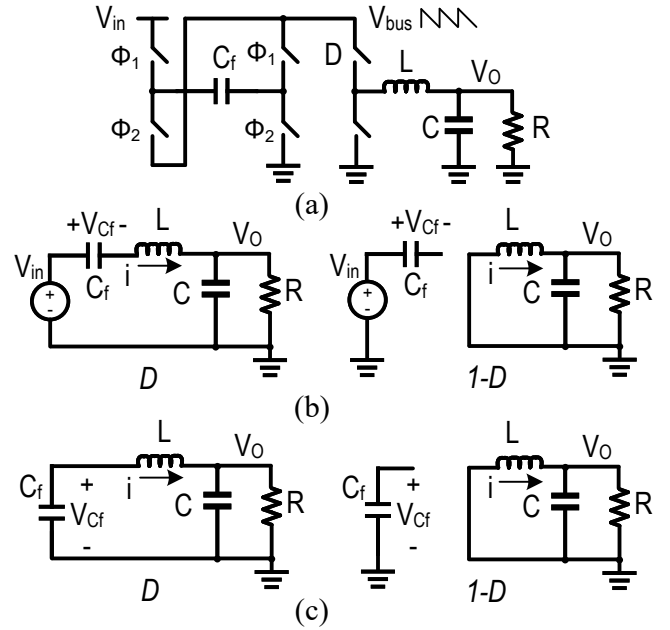


Fig. 7. (a) 1SC + 1Buck circuit schematic and (b) flying capacitor charge state Φ_1 and (c) flying capacitor discharge state Φ_2 during buck converter D and $1 - D$. D is the period when the switch is on.

$$A'_1 = \begin{bmatrix} \frac{-1}{RC} & 0 & \frac{1}{C} \\ 0 & 0 & 0 \\ \frac{-1}{L} & 0 & 0 \end{bmatrix}, \quad B'_1 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Similarly, from Fig. 7 (c) one can obtain the following state matrix A_2 , A'_2 , B_2 , and B'_2 :

$$A_2 = \begin{bmatrix} \frac{-1}{RC} & 0 & \frac{1}{C} \\ 0 & 0 & \frac{1}{C_f} \\ \frac{-1}{L} & \frac{1}{L} & 0 \end{bmatrix}, \quad B_2 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$A'_2 = \begin{bmatrix} \frac{-1}{RC} & 0 & \frac{1}{C} \\ 0 & 0 & 0 \\ \frac{-1}{L} & 0 & 0 \end{bmatrix}, \quad B'_2 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Since the SC charge time and discharge time is always equal to each other $\Phi_1 = \Phi_2 = 0.5T_{sc}$, in average we have the average state vector A

$$A = \frac{D(A_1 + A_2) + (1-D)(A'_1 + A'_2)}{2} = \begin{bmatrix} \frac{-1}{RC} & 0 & \frac{1}{C} \\ 0 & 0 & 0 \\ \frac{-1}{L} & 0 & 0 \end{bmatrix} \quad (5)$$

Similarly, apply same averaging method to B we have

$$B = \begin{bmatrix} 0 \\ 0 \\ \frac{D}{2L} \end{bmatrix} \quad (6)$$

From (5) we learn that the V_{Cf} information is canceled to each other during the calculation. The dc operation point X_0 can be solved by letting (2) equal to 0, which is given as

$$X_0 = -A^{-1}B \quad (7)$$

$$X_0 = \begin{bmatrix} V_O \\ V_{Cf} \\ I_L \end{bmatrix} = \begin{bmatrix} \frac{DV_{in}}{2} \\ \frac{V_{in}^2}{2} \\ \frac{DV_{in}}{2R} \end{bmatrix} \quad (8)$$

Next we do perturbation analysis ($x = X + \hat{x}$, $X \gg \hat{x}$) to (2) and achieves

$$\dot{\hat{x}}(t) = A\hat{x} + B\hat{u} + E\hat{d} \quad (9)$$

where

$$E = (A_1 - A'_1)X_0 + (B_1 - B'_1)U_0 \quad (10)$$

Then the control transfer functions can be derived by assuming $\hat{u} = 0$ in (9)

$$T = \frac{\hat{x}}{\hat{d}} = (sI - A)^{-1}E \quad (11)$$

Finally the control-to-output transfer function G_{vd} is achieved

$$G_{vd} = \frac{\hat{v}_O}{\hat{d}} = \frac{V_{in}}{2(LCs^2 + \frac{L}{R}s + 1)} \quad (12)$$

Compare (12) to the regular buck converter's transfer function, the only difference is the factor $1/2$, which makes sense because of the 2:1 SC in the front end of the 1SC + 1 Buck topology. This verified the modified averaging method in single-output state-space averaging arrangement. The next step is to apply this averaging method to more sophisticated multi-output configurations.

B. 1SC + 1Buck: Dynamic Behavior Analysis

In a Chiplet-LEGO converter, the bus voltage is the input voltage of the back-end regulator and changes periodically. It may significant impact the dynamic behavior of the full system. Let's recall the complete dc-dc converter output voltage in the s domain

$$\hat{v}_{out}(s) = G_{vd}(s)\hat{d}(s) + G_{vg}(s)\hat{v}_{bus}(s) \quad (13)$$

where G_{vg} refer to input-to-output transfer function. $\hat{v}_{out}$ is not linear with respect to $\hat{d}$ when $\hat{v}_{bus}(s)$ is not 0. Fig. 8 (a)

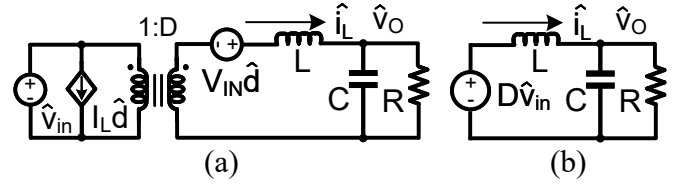


Fig. 8. (a) Small signal model for the buck converter, and (b) when $\hat{d} = 0$.

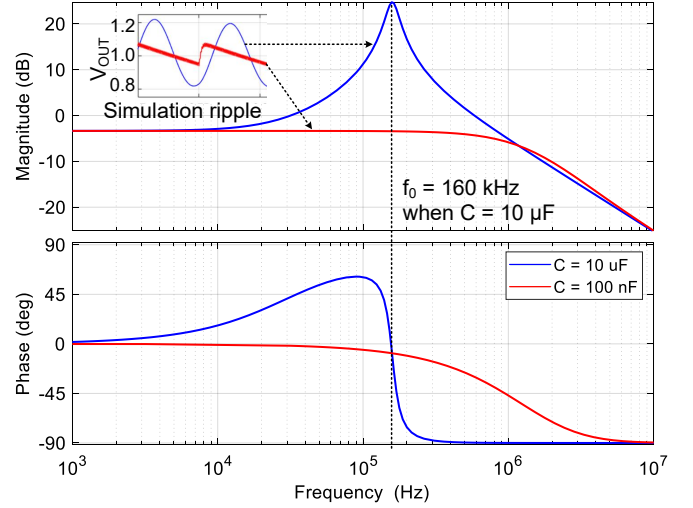


Fig. 9. Bode plot of $G_{v_{in}i_L}$ when buck converter output capacitor $C = 10 \mu F$ and 100 nF .

presents the small signal model of a buck converter and (b) presents the small signal model when $\hat{d} = 0$.

From Fig. 8 (b) one can derive the input-to-inductor current transfer function

$$G_{i_L v_{in}}(s) = \frac{\hat{i}_L}{\hat{v}_{in}} = \frac{D}{R} \frac{(1 + sRC)}{LCs^2 + \frac{L}{R}s + 1} \quad (14)$$

where $\omega_z = \frac{1}{RC}$, $\omega_0 = \frac{1}{\sqrt{LC}}$, $Q = R\sqrt{\frac{C}{L}}$. (14) describes how much inductor current changes with respect to input voltage of a buck converter and Fig. 9 shows its bode plot when $V_{IN} = 6 \text{ V}$, $D = 0.34$, $R = 0.5 \Omega$, $L = 100 \text{ nH}$, and $C_f = 10 \mu F$. A peak resonant gain is observed at the nature frequency $f_0 = 160 \text{ kHz}$ when $C = 10 \mu F$, indicating that the magnitude of i_L is amplified due to the change of v_{in} . However, the gain is always smaller than 0 dB when $C = 1 \mu F$, indicating that we can avoid this amplification if we design buck converter properly according to Q , or design f_{sc} much larger than the nature frequency ω_0 of buck converter

$$2f_{sc} \gg \frac{\omega_0}{2\pi}. \quad (15)$$

The $2f_{sc}$ shown in (15) comes from the fact that the real input voltage variation frequency seen by inductor current is doubled due to the 50% duty ratio in the SC front-end. Figure 9 reveals that despite using a capacitance 100 times larger, the output ripple is amplified due to the resonance aligning with the f_{sc} – a critical consideration in converter design.

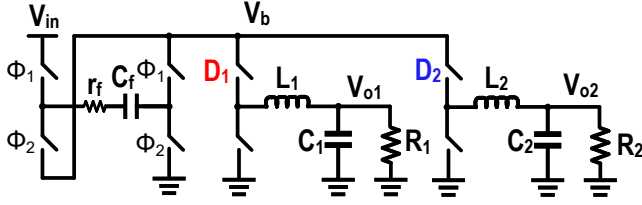


Fig. 10. 1SC + 2Buck converter circuit schematic, where r_f represents the sum of parasitic flying capacitor resistance and switches on-resistance.

C. 1SC + 2Buck: Cross Regulation Transfer Function

In a multi-outputs DC-DC converter, cross-regulation refers to the undesirable change in one output voltage when the load on a different output is changing. The transfer function $G_{v_{o2}d_1} = \hat{v}_{o2}/\hat{d}_1$ is needed to investigate the cross regulation problem of 1SC + 2BUCK topology as shown in Fig. 10. Notice that equivalent resistance r_f is necessary in this circuit modeling because it links between i_{L1} and i_{L2} . (16) and (17) describe the inductor current state-equations during Φ_1 of i_{L1} and i_{L2} , respectively.

$$L \frac{di_{L1}}{dt} = V_{in} - V_{cf} - i_{L1}r_f - i_{L2}r_f - v_{o1} \quad (16)$$

$$L \frac{di_{L2}}{dt} = V_{in} - V_{cf} - i_{L1}r_f - i_{L2}r_f - v_{o2} \quad (17)$$

Define $x = [v_{o1} \ v_{o2} \ v_{cf} \ i_{L1} \ i_{L2}]^T$ and the input vector $u = [v_{in}]$. Similarly, we obtain the state matrix by writing the state equations. For simplicity, we present the calculated result directly. Notice that this analysis is only held when $D_1 > D_2$ and the two buck converters operating at same switching frequency ($f_{b1} = f_{b2}$) without any delay to each other. The transfer function $G_{v_{o2}d_1}$, which is given by

$$G_{v_{o2}d_1} = \frac{\hat{v}_{o2}}{\hat{d}_1} = \frac{\hat{v}_b}{\hat{d}_1} \cdot \frac{\hat{v}_{o2}}{\hat{v}_b} \quad (18)$$

where

$$\frac{\hat{v}_b}{\hat{d}_1} = \frac{I_{L1}r_f(C_1R_1s + 1)}{D_1(L_1C_1s^2 + (C_1D_1r_f + \frac{L_1}{R_1})s + 1 + \frac{D_1r_f}{R_1})} \quad (19)$$

$$\frac{\hat{v}_{o2}}{\hat{v}_b} = \frac{D_2}{(L_2C_2s^2 + (C_2D_2r_f + \frac{L_2}{R_2})s + 1 + \frac{D_2r_f}{R_2})} \quad (20)$$

$$\omega_1 = \sqrt{\frac{R_1 + D_1r_f}{R_1L_1C_1}} \approx \sqrt{\frac{1}{L_1C_1}} \quad (R_1 \gg D_1r_f)$$

$$\omega_2 = \sqrt{\frac{R_2 + D_2r_f}{R_2L_2C_2}} \approx \sqrt{\frac{1}{L_2C_2}} \quad (R_2 \gg D_2r_f)$$

(18) can be regarded as the product of control-to-input transfer function (19) of buck converter 1 and input-to-output transfer function (20) of buck converter 2. Fig. 11 compares the bode plots between SIMPLIS simulation and the calculated transfer function by (18) when $V_{IN} = 12$ V, $V_{O1} = 1$ V ($D_1 = 0.17$), $V_{O2} = 0.6$ V ($D_2 = 0.1$), $R_1 = R_2 = 0.2 \ \Omega$, $r_f = 10 \ m\Omega$, $L_1 = L_2 = 100$ nH, $C_1 = C_2 = C = 10 \ \mu F$ and $100 \ \mu F$. Since the transfer function (18) measures the change of V_{O2} caused by the change of D_1 , indicating that the cross regulation is insignificant when the resonant peak is smaller than 0 dB (theoretically, the gain is smaller the better).

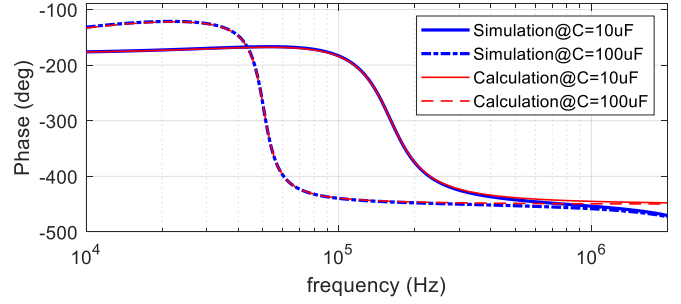
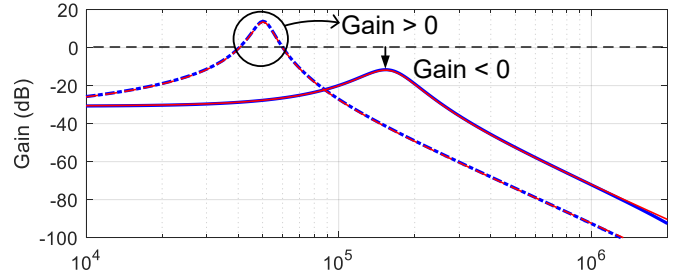


Fig. 11. SIMPLIS simulation and calculated bode plots verifying the deduced transfer function. Gain can be larger than 0 when $C = 100 \ \mu F$ while gain is always smaller than 0 when $C = 10 \ \mu F$.

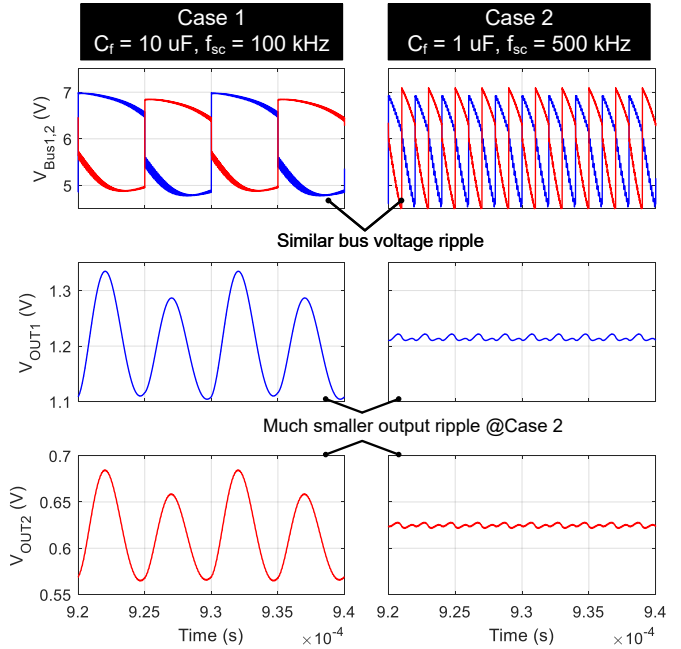


Fig. 12. The two simulation results compare the output voltage ripple when $f_{sc} = 100$ and 500 kHz and $f_0 \approx 200$ kHz.

D. 2SC + 4Buck: Dual-output Chiplet-LEGO Dynamic

Fig. 12 shows the output voltage ripple dynamics of the Chiplet-LEGO. A good f_{sc} selection is needed, based as on (15). When $V_{IN} = 24$ V, $V_{O1} = 1.2$ V, $V_{O2} = 0.6$ V, $R_1 = R_2 = 0.2 \ \Omega$, $r_f = 10 \ m\Omega$, $L_1 = L_2 = 100$ nH, $C = 12 \ \mu F$, and $f_b = 10$ MHz, in Case 1, we have $2f_{sc} \approx f_0 = 200$ kHz and $C_1 = C_2 = C_3 = 10 \ \mu F$, and in Case 2, we have $f_{sc} = 500$ kHz and $C_1 = C_2 = C_3 = 1 \ \mu F$. Although Case 2 exhibits a larger bus voltage ripple than Case 1, it results in

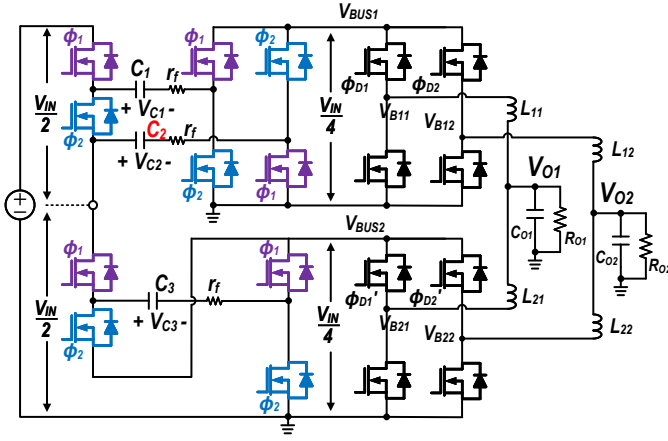


Fig. 13. Circuit schematic of 2SC + 4Buck, including the parasitic resistance r_f which represents the total DCR along the flying capacitor current path.

a significantly smaller output voltage ripple. This is because in Case 1, $2f_{sc} = f_0$ causing the bus voltage ripple to be amplified by the LC resonance as seen from the output.

As illustrated in Fig. 13, there are a total of 9 state variables (3 flying capacitors, 4 inductors and 2 output capacitors) in the state vector $x = [v_{O1}, v_{O2}, v_{c1}, v_{c2}, v_{c3}, i_{L11}, i_{L12}, i_{L21}, i_{L22}]^T$, and the input vector $u = [v_{in}]$, D_a and D_b ($D_a = D_b = D_1$) refer to the duty ratios applied to the buck converters relate to V_{O1} (L_{11} and L_{21}), respectively. The state matrices A_{1a} and B_{1a} in Φ_1 during D_a , A_{2a} and B_{2a} in Φ_2 during D_a , A_{1b} and B_{1b} in Φ_1 during D_b , and A_{2b} and B_{2b} in Φ_2 during D_b relate V_{O1} can be obtained. D_c and D_d ($D_c = D_d = D_2$) refer to the duty ratios applied to the buck converters relate to V_{O2} (L_{12} and L_{22}), respectively. Since D_1 is treated as a control variable, the state-matrix of i_{L3} and i_{L4} can be combined together as a new state-matrix. As a result, the new state matrix A_{1cd} and B_{1cd} in Φ_1 during D_2 , A'_{1cd} and B'_{1cd} in Φ_1 during D'_2 , A_{2cd} and B_{2cd} in Φ_2 during D_2 , A'_{2cd} and B'_{2cd} in Φ_2 during D'_2 relate V_{O2} can be obtained.

Small-signal perturbation analysis is performed by expressing each duty ratio as $d = D + \hat{d}$, $D \gg \hat{d}$. In the interleaved two-phase buck stages, this yields two modes [4]:

- **Common-mode perturbation:** $d_a = d_b = D + \hat{d}$. This mode cancels the flying-capacitor dynamics and governs the output-capacitor current and voltage behavior.
- **Differential-mode perturbation:** $d_a = D + \hat{d}$, $d_b = D - \hat{d}$. This mode cancels the output-capacitor dynamics and governs the flying-capacitor current and voltage behavior.

The common-mode behavior and differential-mode behavior are orthogonal to each other and superposition applies. Under these definitions, the common-mode small-signal state-space matrix E_C is derived in (21), and the differential-mode matrix E_D is derived in (22). Applying (21) to (11) we can obtain the cross regulation transfer function $G_{v_{o2}d_1}$. Since the complete transfer function $G_{v_{o2}d_1}$ is too complex, we denote $G_{v_{o2}d_1,sp}$ as simplified transfer function, which shares the same concept of the previous 1SC + 2Buck converter cross regulation transfer function (18) and is given by

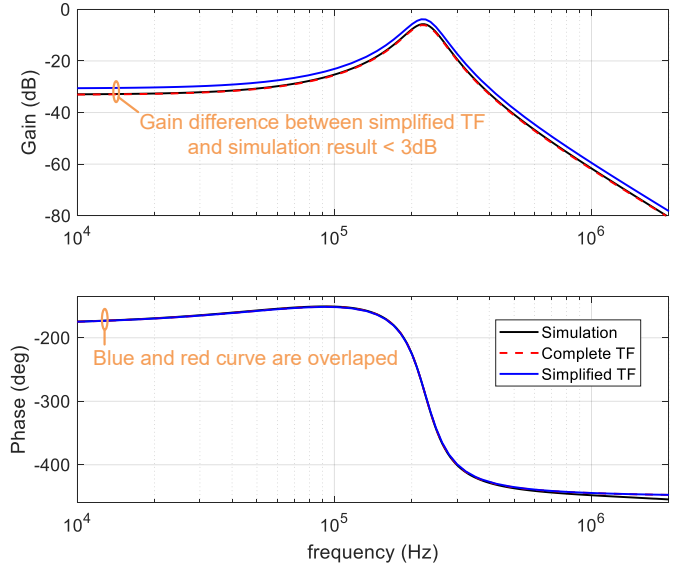


Fig. 14. 2SC + 4buck converter simulation, complete and simplified TF bode plots of cross regulation.

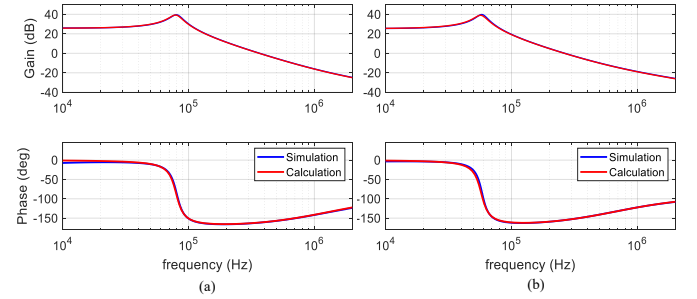


Fig. 15. 2SC + 4buck converter simulation and calculation bode plots of interphase LC response bode plot when (a) $D_1 = 0.4$ and $D_2 = 0.3$, and (b) $D_1 = 0.2$ and $D_2 = 0.3$.

$$G_{v_{o2}d_1,sp} = \frac{\hat{v}_{o2}}{\hat{d}_1} = \frac{\hat{v}_b}{\hat{d}_1} \cdot \frac{\hat{v}_{o2}}{\hat{v}_b} \quad (24)$$

where

$$\frac{\hat{v}_b}{\hat{d}_1} \approx \frac{2I_{L11}r_f(C_1R_1s + 1)}{D_1(L_{1p}C_1s^2 + (C_1D_1r_f + \frac{L_{1p}}{R_1})s + 1 + \frac{D_1r_f}{R_1})} \quad (25)$$

$$\frac{\hat{v}_{o2}}{\hat{v}_b} \approx \frac{D_2}{(L_{2p}C_2s^2 + (C_2D_2r_f + \frac{L_{2p}}{R_2})s + 1 + \frac{D_2r_f}{R_2})} \quad (26)$$

$$\omega_1 = \sqrt{\frac{R_1 + D_1r_f}{R_1L_{1p}C_1}} \approx \sqrt{\frac{1}{L_{1p}C_1}} \quad (R_1 \gg D_1r_f)$$

$$\omega_2 = \sqrt{\frac{R_2 + D_2r_f}{R_2L_{2p}C_2}} \approx \sqrt{\frac{1}{L_{2p}C_2}} \quad (R_2 \gg D_2r_f)$$

The operating conditions are $V_{IN} = 24$ V, $V_{O1} = 1$ V ($D_1 = 0.17$), $V_{O2} = 0.6$ V ($D_2 = 0.1$), $R_1 = R_2 = 0.2$ Ω , $r_f = 10$ m Ω , $L_{1p} = L_{2p} = \frac{L_1}{2} = \frac{L_2}{2} = 50$ nH, $C_1 = C_2 = C = 10$ μ F. Applying (22) to (11) we obtain the interphase control-to-flying capacitor transfer function $G_{v_{c2}d_1}$, which is given as (23). Fig. 15 illustrates the simulation and calculated transfer function matching each other when frequency is higher than

$$E_C = \frac{((A_{1a} + A_{2a} - A'_{1a} - A'_{2a}) + (A_{1b} + A_{2b} - A'_{1b} - A'_{2b}))X_0 + ((B_{1a} + B_{2a} - B'_{1a} - B'_{2a}) + (B_{1b} + B_{2b} - B'_{1b} - B'_{2b}))U_0}{2} \quad (21)$$

$$E_D = \frac{((A_{1a} + A_{2a} - A'_{1a} - A'_{2a}) - (A_{1b} + A_{2b} - A'_{1b} - A'_{2b}))X_0 + ((B_{1a} + B_{2a} - B'_{1a} - B'_{2a}) - (B_{1b} + B_{2b} - B'_{1b} - B'_{2b}))U_0}{2} \quad (22)$$

$$G_{v_{c2d1}} = \frac{\hat{v}_{C2}}{\hat{d}_1} \approx \frac{D_1 V_{IN} (4R_1 + 3D_1 r_f + 3D_2 r_f + 2Ls)}{2(4R_1 + 3D_1 r_f + 3D_2 r_f)(2C_2 L s^2 + (3C_2 r_f(D_1 + D_2))s + D_1^2 + D_2^2)} \quad (23)$$

$$\omega_z = \frac{4R_1 + 3r_f(D_1 + D_2)}{2L}, \quad \omega_0 = \sqrt{\frac{D_1^2 + D_2^2}{2C_2 L}}, \quad Q = \frac{2}{3r_f(D_1 + D_2)} \sqrt{\frac{(D_1^2 + D_2^2)L}{2C_2}}$$

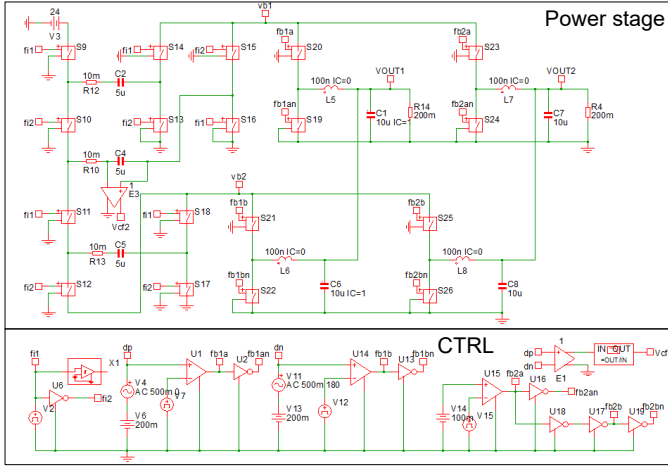


Fig. 16. Schematic of the Chiplet-LEGO simulation platform in Simplis.

10 kHz. Fig. 15 (a) shows the case of $D_1 > D_2$ and (b) shows the case of $D_2 > D_1$. The simulation conditions are $V_{IN} = 24$ V, $R_1 = R_2 = 0.2 \Omega$, $r_f = 10 \text{ m}\Omega$, $L_1 = L_2 = 100 \text{ nH}$, $C_1 = C_2 = C_3 = 5 \mu\text{F}$, and $C_{O1} = C_{O2} = 10 \mu\text{F}$. Fig. 16 presents the Chiplet-LEGO simulation platform in Simplis.

V. CONCLUSION

This paper presents the Chiplet-LEGO VRM architecture, a promising candidate for delivering power to future high current chiplets with scalability and flexibility. A 24 V-to-1 V dual-output Chiplet-LEGO prototype is demonstrated, achieving a peak efficiency of 85.8% with two independent 0.5-1.2 V output rails at a 120 W rated power. The Chiplet-LEGO architecture offers high modularity via shared voltage buses and extendable power ratings. Comprehensive modeling of the system's dynamic transients was developed to provide useful guidance for operating the Chiplet-LEGO architecture.

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